

Exact Polynomial Synthesis from Rvachev Up-Atoms

Common-scale dictionaries, antiderivative-train windows,
Thue–Morse certificates, and minimal atom counts

Consolidated frontier volume for Vladimir Reshetnikov’s ProveIt project

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Abstract

Let $u = \text{up}$ be Rvachev’s even C^∞ probability density supported on $[-1, 1]$. This volume consolidates two independently written reports on the same synthesis problem: represent an arbitrary polynomial *exactly*, on a prescribed compact interval, by a finite signed sum of shifted and scaled copies of u . The two sources solve it by two genuinely different constructions, which are presented here side by side, reconciled, and combined.

The common-scale dictionary. For degree d , the minimal integer scale-to-shift ratio with Strang–Fix reproduction order d is $M = 2^d$. Moment deconvolution $Q = M_u(MD)^{-1}P$ gives a global cardinal identity; on one cell its $2M$ active atoms collapse, because the complete dyadic alias nullspace is governed by the q -integer/cyclotomic annihilator $A_d(z) = \prod_{m=1}^d [2^m]_z$ of degree $2M - d - 2$. The one-cell up-spline space therefore has exact dimension $d + 2$; a proved endpoint basis represents every $P \in \Pi_d$ by $d + 2$ common-scale atoms; no fixed dictionary of nontrivial affine up-atoms spanning Π_d can do better; and an output-sensitive algorithm with Bernoulli differential symbol $H_d(z) = (z/2 \sinh(z/2))^d / M_u(z)$ computes the coefficients in $\mathcal{O}(d^2)$ exact operations. Polynomiality of an arbitrary local up-spline is decided by a *single scalar*, which after one finite difference becomes a signed Thue–Morse block sum; the one-dimensional nonpolynomial defect has an explicit odd-frequency Fourier residue.

The scale-ladder windows. The refinement equation inverts to an antiderivative train $Ju(x) = \sum_{k \geq 0} u((x - 2k - 1)/2)$, whose m -fold iterate has restricted binary partition numbers $p_m(N)$ as weights – the causal Green sequence of the finite Thue–Morse difference operator. Truncating by a support-edge invariance yields exact polynomial windows $\Psi_{n,h}$ at scales $2^{n+1}h$: with $h = (b-a)/2$, every degree uses exactly *two* atoms, so $\deg \leq d$ costs at most $2d + 2$ atoms; smaller h trades atoms for a prescribed support collar. The coefficients are a triangular Appell transform with Bernoulli–Bell closed forms, the block basis is an extended complete Chebyshev system with Vandermonde interpolation determinants, and the construction is an explicit compactly supported C^∞ extension operator for polynomial data.

Combined results. The dictionary bound sharpens the ladder count: the minimal number N_d of up-atoms sufficient for all of Π_d on an interval satisfies $N_0 = 2$ and $N_d \leq d + 2$, with $N_d = d + 2$ conjectured. And the canonical defect is identified, with an explicit constant, as the first-failure trigonometric series of the shifted dyadic-comb quadrature of the companion volume *Dyadic-Comb Frontiers*: its special values are governed by the spectral Dirichlet values $D(2r)$ proved there, giving the exact rational evaluations

$$\mathcal{E}_1(0) = \frac{1}{18}, \quad \mathcal{E}_3(0) = -\frac{23}{1350}, \quad \mathcal{E}_5(0) = \frac{1543}{119070}, \quad \mathcal{E}_7(0) = -\frac{1196113}{65063250},$$

two of which independently reconfirm exact phase-error values computed in that volume. Provenance of the absorbed sources, with SHA-256 hashes, is in [section B](#).

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Provenance and merge conventions

This volume is an editorial consolidation (28 August 2026) of two fourth-wave drafts that solve the same problem by different routes; shared foundations are stated once, each construction keeps its own chapter, their overlapping material (Appell calculus, Thue–Morse products, rationality, tensor products, extension operators, conditioning, Lean roadmaps) is merged, and the comparison chapter and the cross-volume defect identities are new to the consolidation. [Section B](#) records the absorbed sources with SHA-256 hashes and the deduplication detail; their supporting files are under `assets/`.

Status discipline

Theorems, propositions, lemmas, and corollaries have complete proofs here or explicit citations to the repository’s primary exposition. The classical fact that Rvachev’s atomic-function spaces contain polynomials goes back to V. L. and V. A. Rvachev [3, 5]; novelty claims concern the explicit finite constructions, dimensions, algorithms, certificates, and identities below, relative to the audited corpus and a targeted literature search — never unqualified global priority. Conjectures and problems are labeled. Nothing here is formalized in Lean yet.

Notation

u = up as in the corpus; $\widehat{f}(\xi) = \int f e^{-2\pi i \xi x} dx$; $\widehat{u}(\xi) = \prod_{j \geq 0} \text{sinc}_\pi(\xi/2^j)$ with $\text{ord}_{\xi=n} \widehat{u} = 1 + \nu_2(n)$ at nonzero integers n . The probability model is $Z = \sum_{j \geq 1} 2^{-j} V_j$, V_j i.i.d. uniform on $[-1, 1]$, with density u , moment generating function $M_u(z) = \mathbb{E} e^{zZ} = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}$, even moments μ_{2r} ($\mu_2 = \frac{1}{9}$, $\mu_4 = \frac{19}{675}$, $\mu_6 = \frac{583}{59535}$, ...), and cumulants $\kappa_{2m} = B_{2m}/(2m(1 - 4^{-m}))$. Two dual Appell families recur:

$$G_n(y) := \mathbb{E} (y - Z)^n = M_u(D) y^n = \sum_r \binom{n}{r} \mu_r y^{n-r}, \quad \mathcal{A}_n(y) := M_u(D)^{-1} y^n = \sum_r \binom{n}{r} \gamma_r y^{n-r}, \quad (0.1)$$

the *moment family* G_n (the same polynomials as the Appell kernel of the companion volume’s dyadic evaluator) and the *deconvolution family* $\mathcal{A}_n$ (the corpus’s Rvachev–Appell polynomials), where $1/M_u(z) = \sum_r \gamma_r z^r / r!$ with

$$\gamma_2 = -\frac{1}{9}, \quad \gamma_4 = \frac{31}{675}, \quad \gamma_6 = -\frac{2347}{59535}, \quad \gamma_8 = \frac{1904369}{32531625}, \quad \gamma_{10} = -\frac{3309760579}{24405225075}, \quad (0.2)$$

odd coefficients zero, and $\gamma_r = Y_r(-\kappa_1, \dots, -\kappa_r)$ (complete Bell polynomials in the negated cumulants). Thue–Morse signs are $\varepsilon_r = (-1)^{s_2(r)}$, the finite product is $\Theta_m(z) = \prod_{j < m} (1 - z^{2^j}) = \sum_{r < 2^m} \varepsilon_r z^r$, and $p_m(N)$ denotes the restricted binary partition numbers (2.3). Π_d is the space of real polynomials of degree at most d , and $[q]_z = (1 - z^q)/(1 - z)$ the q -integer.

1 The synthesis problem and its boundary conditions

Given $p \in \mathbb{R}[x]$ of degree at most d and a nondegenerate interval $[a, b]$, we seek a *finite* functional identity

$$p(x) = \sum_{r=1}^N c_r u\left(\frac{x - \tau_r}{\sigma_r}\right), \quad x \in [a, b], \quad (1.1)$$

with real amplitudes, shifts, and nonzero scales. Three structural facts frame every construction.

(i) *Globality is impossible; locality is the right question.* Every finite sum (1.1) is compactly supported, so no nonzero polynomial admits a global identity on $\mathbb{R}$; the achievable object is an exact *window* — polynomial on $[a, b]$, smoothly decaying to zero outside a finite support.

(ii) *Signed amplitudes are essential and two atoms are needed even for constants.* By the refinement derivative $u'(x) = 2u(2x+1) - 2u(2x-1)$, the function u is strictly increasing on $(-1, 0)$ and strictly decreasing on $(0, 1)$; hence a single atom is never constant on a nondegenerate interval, and $N \geq 2$ already for $p \equiv 1$. The degree-zero solution is the two-shift partition of unity

$$u\left(\frac{x-a}{L}\right) + u\left(\frac{x-b}{L}\right) = 1, \quad a \leq x \leq b, \quad L = b - a. \quad (1.2)$$

(iii) *One fixed scale cannot reproduce high degree globally.* Under the corpus convention the dual-lattice zeros of a single integer-shift system generated by u have minimum multiplicity one (at odd frequencies), so its global Strang–Fix reproduction order is zero beyond constants [8, 10]: any exact construction must either enlarge the scale-to-shift ratio (the dictionary route of section 3) or use several scales and boundary truncation (the ladder route of section 5).

Within the corpus, the nonduplication boundary is sharp: the audited inverse/sampling volume proves the *opposite orientation*

$$x^n = 2^{-N} \sum_{k \in \mathbb{Z}} \mathcal{A}_n\left(x - \frac{k+\theta}{2^N}\right) u\left(\frac{k+\theta}{2^N}\right) \quad (n \leq N), \quad (1.3)$$

where the translated functions are *polynomials* and the up-values are scalar weights. Here the translated and dilated copies of u are the functions and polynomial data become their amplitudes. Two other imported facts do heavy work below: the exact integer zero orders of $\hat{u}$, and the corpus’s non-elementarity theorem, which implies that no nonzero affine copy of u agrees with a polynomial on any open subinterval of the interior of its support — the fact that converts dimension *bounds* into dimension *equalities*.

2 Shared calculus: Appell pairs, Thue–Morse products, and binary partitions

2.1 The Appell pair

The two families (0.1) are formal inverses: $M_u(D)\mathcal{A}_n = y^n$ and $M_u(D)^{-1}G_n = y^n$, both operators triangular with unit diagonal on polynomials. Their generating functions are $\sum_n G_n(y)t^n/n! = e^{yt}M_u(t)$ and $\sum_n \mathcal{A}_n(y)t^n/n! = e^{yt}/M_u(t)$; both satisfy the Appell derivative ladder $G'_n = nG_{n-1}$, $\mathcal{A}'_n = n\mathcal{A}_{n-1}$; and their coefficient sequences (μ_r) , (γ_r) are mutually inverse under binomial convolution, with

$$\gamma_0 = 1, \quad \gamma_n = - \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \gamma_{n-r}, \quad \mu_n = \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \mu_{n-r}. \quad (2.1)$$

The first moment family members are $G_0 = 1$, $G_1 = y$, $G_2 = y^2 + \frac{1}{9}$, $G_3 = y^3 + \frac{1}{3}y$, $G_4 = y^4 + \frac{2}{3}y^2 + \frac{19}{675}$, $G_5 = y^5 + \frac{10}{9}y^3 + \frac{19}{135}y$.

2.2 Thue–Morse products, q -integers, and their Green sequences

Three finite dyadic products organize everything discrete in this volume:

$$\Theta_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}), \quad A_d(z) = \prod_{m=1}^d [2^m]_z = \prod_{m=1}^d \frac{1 - z^{2^m}}{1 - z}, \quad \frac{1}{\Theta_m(z)} = \sum_{N \geq 0} p_m(N) z^N. \quad (2.2)$$

Definition 2.1 (Restricted binary partitions). $p_m(N)$ counts representations $N = \sum_{j=0}^{m-1} 2^j k_j$ with $k_j \in \mathbb{N}_0$; equivalently (2.2) holds, and

$$p_m(N) = p_{m-1}(N) + p_m(N - 2^{m-1}), \quad p_m(N < 0) = 0, \quad p_m(0) = p_m(1) = 1. \quad (2.3)$$

Proposition 2.2 (Green pair and quasipolynomial structure). *The partition sequence is the causal convolution inverse of the signed Thue–Morse block: $\sum_r \varepsilon_r p_m(N-r) = \delta_{N0}$, i.e. $[\prod_{j < m} (I - S^{2^j})] p_m = \delta_0$ for the backward shift S . Moreover $p_m(N)$ is a quasipolynomial in N of degree $m-1$ and period dividing 2^{m-1} , with residue-independent leading term*

$$p_m(N) = \frac{N^{m-1}}{(m-1)! 2^{m(m-1)/2}} + \mathcal{O}(N^{m-2}); \quad (2.4)$$

hence the normalized weights $w_{n,N} = n! 2^{n(n+1)/2} p_{n+1}(N)$ are monic degree- n quasipolynomials, $w_{n,N} = N^n + \mathcal{O}(N^{n-1})$.

Proof. The Green identity is $\Theta_m(z) \cdot \Theta_m(z)^{-1} = 1$ read coefficientwise. The generating function is rational with all poles at 2^{m-1} -th roots of unity, giving the quasipolynomial form; the pole at $z = 1$ has order m with $\Theta_m(z) = 2^{m(m-1)/2} (1-z)^m (1 + \mathcal{O}(1-z))$, whose $(1-z)^{-m}$ coefficient $\binom{N+m-1}{m-1}$ gives (2.4). $\square$

The annihilator A_d is the mirror object: where Θ^{-1} integrates m dyadic differences (the ladder construction), A_d applies them after removing the pole (the dictionary construction). Its structure theorem:

Theorem 2.3 (q -integer and cyclotomic factorization). *With $M = 2^d$ and $\omega = e^{2\pi i/M}$,*

$$A_d(z) = \prod_{j=1}^{M-1} (z - \omega^j)^{\nu_2(j)+1} = \prod_{\ell=1}^d (z^{2^{\ell-1}} + 1)^{d-\ell+1}, \quad (2.5)$$

monic, palindromic, with positive integer coefficients, $\deg A_d = 2M - d - 2$, and $A_d(1) = 2^{d(d+1)/2}$. Writing $A_d(z) = \sum_{\ell} a_{\ell} z^{\ell}$, the coefficient a_{ℓ} counts tuples $(r_1, \dots, r_d)$ with $0 \leq r_m < 2^m$ and $\sum r_m = \ell$; in particular (a_{ℓ}) is symmetric, unimodal, and log-concave, and $a_{\ell}/A_d(1)$ is the law of a sum $L = \sum_{m=1}^d U_m$ of independent U_m uniform on $\{0, \dots, 2^m - 1\}$.

Proof. The roots of $z^{2^{\ell-1}} + 1$ are the ω^j with $\nu_2(j) = d - \ell$, required with multiplicity $d - \ell + 1$; since $[2^m]_z = \prod_{\ell \leq m} (z^{2^{\ell-1}} + 1)$, the multiplicities telescope to (2.5). Degree and mass follow by summing $2^m - 1$ and multiplying 2^m ; the coefficient statement is the expansion of the product of uniform blocks, each log-concave, and convolution preserves log-concavity. $\square$

Finally, the two products meet in one factorization that will become the polynomiality certificate:

$$A_d(z) (z - 1)^{d+1} = (-1)^{d+1} (1 - z) \Theta_d(z^2) = (-1)^{d+1} (1 - z) \sum_{r=0}^{M-1} \varepsilon_r z^{2^r}. \quad (2.6)$$

3 The common-scale dictionary

The first construction fixes *one* dyadic scale and lets the alias arithmetic of the sinc product do all the work. Normalize atoms by

$$\psi_{\rho,k}(t) = \frac{1}{\rho} u\left(\frac{t-k}{\rho}\right), \quad \widehat{\psi_{\rho,k}}(\xi) = e^{-2\pi i k \xi} \widehat{u}(\rho \xi), \quad (3.1)$$

supported on $[k - \rho, k + \rho]$ with unit integral.

3.1 Why the minimal cardinal ratio is 2^d

Theorem 3.1 (Scale classification). *The integer translates $\{\psi_{\rho,k}\}_{k \in \mathbb{Z}}$ form a partition of unity iff $\rho \in \mathbb{N}$; for integer ρ the Strang–Fix polynomial reproduction order of the system is exactly $\nu_2(\rho)$. Hence the smallest ratio reproducing all of Π_d is $\rho = 2^d$, and odd oversampling ($\rho = 2^d q$, q odd) buys no order.*

Proof. The n -th Fourier coefficient of the periodized sum is $\widehat{u}(\rho n)$, which vanishes for all $n \neq 0$ iff ρn is always a nonzero integer, i.e. $\rho \in \mathbb{N}$. For integer ρ and $n \neq 0$, $\text{ord}_{\xi=n} \widehat{u}(\rho \cdot) = 1 + \nu_2(\rho) + \nu_2(n)$, minimal at odd n ; the classical Strang–Fix equivalence [8, 9] gives order $\nu_2(\rho)$. $\square$

Fix henceforth $M = 2^d$ and $\psi_k = \psi_{M,k}$.

3.2 Global synthesis by moment deconvolution

Lemma 3.2 (Polynomial-coefficient Poisson collapse). *For every $Q \in \Pi_d$,*

$$\sum_{k \in \mathbb{Z}} Q(k) \psi_k(t) = \int_{-1}^1 Q(t - My) u(y) dy = M_u(MD) Q(t), \quad (3.2)$$

a locally finite sum.

Proof. Apply Poisson summation to $x \mapsto Q(x) \psi_0(t - x + \cdot)$ – i.e. to $f_t(x) = Q(x) \frac{1}{M} u((t - x)/M)$, which is C_c^∞ . The zero-frequency term is the stated average. Each nonzero Fourier sample $\widehat{f}_t(n)$ is a combination of derivatives of $\widehat{u}(M\xi)$ through order d at $\xi = n$, and $\text{ord} = 1 + \nu_2(Mn) = d + 1 + \nu_2(n) > d$, so every alias vanishes. Evenness of u turns the average into $M_u(MD)Q$. $\square$

Theorem 3.3 (Global exact up-atom synthesis). *For $P \in \Pi_d$ let*

$$Q = M_u(MD)^{-1} P = P - \frac{M^2}{18} P'' + \frac{31M^4}{16200} P^{(4)} - \frac{2347M^6}{42865200} P^{(6)} + \dots = \sum_{n=0}^d \gamma_n M^n \frac{P^{(n)}}{n!}, \quad (3.3)$$

a terminating series. Then

$$P(t) = \frac{1}{M} \sum_{k \in \mathbb{Z}} Q(k) u\left(\frac{t - k}{M}\right), \quad t \in \mathbb{R}, \quad (3.4)$$

and for $P(t) = t^n$ the coefficient polynomial is the scaled deconvolution Appell polynomial $Q(t) = M^n \mathcal{A}_n(t/M)$.

Restricted to one normalized cell $[0, 1]$, only the $2M$ centers $-M + 1 \leq k \leq M$ act. A first compression is the *partition pair* identity $\psi_{r-M} + \psi_r = \frac{1}{M}$ on $[0, 1]$ ($1 \leq r \leq M$), which folds the $2M$ atoms to $M + 1$; but the true dimension is exponentially smaller.

3.3 The alias nullspace and the local dimension theorem

Theorem 3.4 (Complete dyadic alias family). *Let $\omega = e^{2\pi i/M}$. For every $1 \leq j < M$ and $0 \leq s \leq \nu_2(j)$,*

$$\sum_{k \in \mathbb{Z}} k^s \omega^{jk} \psi_k(t) = 0 \quad (t \in \mathbb{R}), \quad (3.5)$$

and these $R_d := \sum_{j=1}^{M-1} (\nu_2(j) + 1) = 2M - d - 2$ relations are linearly independent on any R_d consecutive integers. The solution space of the recurrence $A_d(E)n = 0$ (shift E) is exactly their span, with A_d the annihilator of [theorem 2.3](#).

Proof. The Fourier transform of the twisted comb $\sum_k k^s \omega^{jk} \delta_k$ is carried, with derivatives through order s , by the coset $j/M + \mathbb{Z}$; there the multiplier $\widehat{u}(M\xi)$ vanishes to order $1 + \nu_2(j + Mn) = 1 + \nu_2(j) > s$, since $0 < j < M$. Independence is the confluent Vandermonde theorem for the distinct roots ω^j with multiplicities $\nu_2(j) + 1$; the count is $\sum_{r < d} 2^{d-r-1}(r+1) = 2M - d - 2$. $\square$

Theorem 3.5 (Exact local dimension). *Let $V_d = \text{span}\{\psi_k|_{[0,1]} : -M+1 \leq k \leq M\}$. Then*

$$\boxed{\dim V_d = d + 2, \quad V_d \cap \mathbb{R}[t] = \Pi_d.} \quad (3.6)$$

The kernel of the synthesis map $(q_k) \mapsto \sum q_k \psi_k|_{[0,1]}$ is exactly the restriction of the A_d -recurrence solution space.

Proof. The R_d relations give $\dim V_d \leq 2M - R_d = d + 2$; [theorem 3.3](#) gives $\Pi_d \subset V_d$, so $\dim V_d \geq d + 1$; and since some active atom (e.g. ψ_0) is not polynomial on $(0, 1)$ — the non-elementarity import of [section 1](#) — the containment is strict, forcing $d + 2$. If V_d contained a polynomial outside Π_d , then $V_d \subseteq \mathbb{R}[t]$ would make every atom polynomial, the same contradiction; and a kernel larger than the recurrence space would push $\dim V_d$ below $d + 2$. $\square$

Proposition 3.6 (Universal fixed-dictionary lower bound). *If finitely many nontrivial affine up-atoms $f_r(x) = u((x - \tau_r)/\sigma_r)$ span a space containing Π_d on a nondegenerate interval I , then their number is at least $d + 2$. Consequently a generic $P \in \Pi_d$ cannot be represented by $d + 1$ or fewer atoms of the dyadic dictionary.*

Proof. No $f_r|_I$ is polynomial (same non-elementarity fact). With $N \leq d$ the dimension is insufficient; with $N = d + 1$, containment of the $(d + 1)$ -dimensional $\Pi_d|_I$ forces equality of spans, making every $f_r|_I$ polynomial — impossible. For genericity, the union over all $\leq (d + 1)$ -subsets of (proper) subspaces $\text{span}\{f_r\} \cap \Pi_d$ is a finite union of proper subspaces, of measure zero in coefficient space. $\square$

3.4 The endpoint basis

Theorem 3.7 (Unique $(d + 2)$ -atom endpoint synthesis). *The atoms with the $d + 2$ rightmost centers, $\mathcal{B}_d = \{\psi_k|_{[0,1]} : M - d - 1 \leq k \leq M\}$, form a basis of V_d . Explicitly: extend the first R_d values $q_k = Q(k)$, $k = -M + 1, \dots, -M + R_d$, by the monic integer recurrence $A_d(E)n = 0$, and put $b_k = q_k - n_k$ on the last $d + 2$ centers; then*

$$\boxed{P(t) = \frac{1}{M} \sum_{k=M-d-1}^M b_k u\left(\frac{t-k}{M}\right), \quad 0 \leq t \leq 1,} \quad (3.7)$$

uniquely in this basis, with rational b_k for rational P . Together with [theorem 3.6](#), $d + 2$ is the least size of any fixed affine up-dictionary spanning Π_d on an interval.

Proof. The extension n lies in the complete nullspace, so its synthesis vanishes; $q - n$ is supported on the last $2M - R_d = d + 2$ centers, so $\mathcal{B}_d$ spans V_d , and [theorem 3.5](#) makes it a basis. Integer recurrence coefficients preserve rationality. Reflection $t \mapsto 1 - t$ gives the mirror-image left basis. $\square$

The first monomial instances (coefficients multiply raw atoms $u((t - k)/M)$):

$$1 = u(t) + u(t - 1), \quad (3.8)$$

$$t = -\frac{1}{2}u\left(\frac{t}{2}\right) + u\left(\frac{t-1}{2}\right) + \frac{1}{2}u\left(\frac{t-2}{2}\right), \quad (3.9)$$

$$t^2 = \frac{13}{9}u\left(\frac{t-1}{4}\right) - \frac{31}{9}u\left(\frac{t-2}{4}\right) + \frac{49}{9}u\left(\frac{t-3}{4}\right) + \frac{5}{9}u\left(\frac{t-4}{4}\right), \quad (3.10)$$

$$t^3 = -14u\left(\frac{t-4}{8}\right) + \frac{136}{3}u\left(\frac{t-5}{8}\right) - \frac{208}{3}u\left(\frac{t-6}{8}\right) + \frac{280}{3}u\left(\frac{t-7}{8}\right) - \frac{106}{3}u\left(\frac{t-8}{8}\right), \quad (3.11)$$

all on $[0, 1]$; the six coefficients for t^4 at scale 16 are $\left(\frac{257824}{675}, -\frac{358624}{225}, \frac{2010496}{675}, -\frac{2528896}{675}, \frac{661024}{135}, -\frac{2567968}{675}\right)$ at centers 11, $\dots$, 16.

3.5 The output-sensitive algorithm

The recurrence route touches $R_d \asymp 2^{d+1}$ values to produce $d + 2$ numbers. The exponential detour can be eliminated entirely: the annihilator acts on the deconvolution sequence as a *bounded* differential operator.

Theorem 3.8 (Compressed residual identity). *Define the Bernoulli differential symbol*

$$H_d(z) = \left(\frac{z}{2 \sinh(z/2)} \right)^d \frac{1}{M_u(z)}, \quad \log H_d(z) = - \sum_{m \geq 1} \frac{B_{2m}}{2m (2m)!} \left(d + \frac{1}{1 - 4^{-m}} \right) z^{2m}. \quad (3.12)$$

Then, with $Q = M_u(MD)^{-1}P$, $K = -M + 1$, and $g_r = (A_d(E)Q)(K + r)$,

$$\boxed{g_r = A_d(1) [H_d(D)P] \left(r - \frac{d}{2} \right) \quad (r \in \mathbb{Z}).} \quad (3.13)$$

Proof. By [theorem 2.3](#), $a_\ell/A_d(1)$ is the law of $L = \sum_m U_m$, so $(A_d(E)Q)(x) = A_d(1) \mathbb{E} Q(x + L)$. Centering L at its mean $R_d/2$ and using $K + R_d/2 = -d/2$,

$$g_r = A_d(1) \left[\mathbb{E} e^{(L - R_d/2)D} M_u(MD)^{-1} P \right] \left(r - \frac{d}{2} \right),$$

and the two product formulas $\mathbb{E} e^{z(L - R_d/2)} = \prod_{m=1}^d \frac{\sinh(2^{m-1}z)}{2^m \sinh(z/2)}$ and $M_u(Mz) = M_u(z) \prod_{r=0}^{d-1} \frac{\sinh(2^r z)}{2^r z}$ cancel factor by factor to (3.12). The logarithm follows from $\log[2 \sinh(z/2)/z] = \sum_m B_{2m} z^{2m}/(2m (2m)!)$ and the cumulant series of M_u . $\square$

Algorithm 3.9 (Output-sensitive exact synthesis). Given $P \in \Pi_d$: truncate H_d at degree $\deg P$ and form $G = H_d(D)P$; compute only $a_0, \dots, a_{d+1}$ of A_d by truncated sliding-window convolution of the uniform blocks; set $g_r = A_d(1) G(r - d/2)$ for $0 \leq r \leq d + 1$; solve the unit-diagonal triangular system $g_r = \sum_{s \leq r} a_{r-s} b_s$ forward; output (3.7). Total cost: $\mathcal{O}(d^2)$ exact rational operations and $\mathcal{O}(d)$ stored rationals – polynomial in the output, not exponential in the scale. On J cells the same procedure with $r \leq J + d$ yields the multi-cell basis below at cost $\mathcal{O}(d^2 + (J + d)^2)$.

For $P \in \mathbb{Z}[t]$ the denominators of all b_k divide $\text{lcm}(\text{den } \gamma_0, \dots, \text{den } \gamma_d)$ (the recurrence is integral), and the raw amplitudes b_k/M add only the dyadic factor.

3.6 Several cells: localization knob

Theorem 3.10 (Multi-cell dimension and basis). *For $J \geq 1$, the space $V_{d,J} = \text{span}\{\psi_k|_{[0,J]} : -M + 1 \leq k \leq J + M - 1\}$ has*

$$\dim V_{d,J} = J + d + 1, \quad V_{d,J} \cap \mathbb{R}[t] = \Pi_d, \quad (3.14)$$

with the $J + d + 1$ consecutive centers $M - d - 1, \dots, M + J - 1$ as a basis. On a physical interval $[a, b]$ with $h = (b - a)/J$ and $P(t) = p(a + ht)$,

$$p(x) = \frac{1}{M} \sum_{r=0}^{J+d} b_r u \left(\frac{x - a - h(M - d - 1 + r)}{Mh} \right), \quad a \leq x \leq b, \quad (3.15)$$

with support footprint $[a - (d + 1)h, b + (2M - 1)h]$. Larger J buys a smaller physical scale $Mh = 2^d(b - a)/J$ at one extra atom per cell.

Proof. The same R_d global relations stay independent, so $\dim \leq J + 2M - 1 - R_d = J + d + 1$; the first-cell endpoint basis plus the atoms centered at $M + 1, \dots, M + J - 1$ are triangularly independent (each new atom vanishes on $[0, r]$ and not immediately to the right), giving equality. A polynomial in $V_{d,J}$ restricts on the first cell into $V_d \cap \mathbb{R}[t] = \Pi_d$. Recurrence elimination leaves exactly the stated centers; the affine substitution gives (3.15). $\square$

4 The polynomiality certificate and the canonical defect

The one-cell space exceeds Π_d by exactly one dimension. Both the membership test for that codimension and the extra direction itself are completely explicit — and both connect this volume to the spectral arithmetic of the companion volume *Dyadic-Comb Frontiers*.

4.1 One scalar decides polynomiality

Theorem 4.1 (Exact certificate, Thue–Morse form). *Order the active coefficients $q = (q_0, \dots, q_{2M-1})$ by increasing center (q_j multiplies ψ_{-M+1+j}), and set*

$$\Lambda_d(q) = [A_d(E) \Delta^{d+1} q]_{-M+1} \stackrel{(2.6)}{=} (-1)^{d+1} \sum_{r=0}^{M-1} \varepsilon_r (q_{2r} - q_{2r+1}). \quad (4.1)$$

Then $\sum_j q_j \psi_{-M+1+j}$ is polynomial on $[0, 1]$ iff $\Lambda_d(q) = 0$; and for the test sequence $q_j = (-M + 1 + j)^{d+1}$, $\Lambda_d(q) = A_d(1) (d + 1)!$.

Proof. Polynomial syntheses are precisely (deconvolution polynomial sequences of degree $\leq d$) + (null sequences): Δ^{d+1} kills the former and $A_d(E)$ the latter, so Λ_d vanishes on the $(2M - 1)$ -dimensional polynomial preimage. It is nonzero on the test sequence by $\Delta^{d+1} k^{d+1} = (d + 1)!$ and $A_d(E)1 = A_d(1)$, so its kernel is exactly that preimage. The Thue–Morse form is the factorization (2.6) applied to the composite operator. $\square$

Thue–Morse bridge

The same finite Prouhet product that annihilates low polynomial moments (Part I of the companion volume) is the complete codimension-one obstruction to a local up-spline being polynomial: the q -integer alias annihilator and the signed Thue–Morse first-difference filter are two factorizations of one operator.

4.2 The canonical defect and its Fourier residue

Set $m = d + 1$ and split the (locally finite) power synthesis $S_m(t) = \sum_k k^m \psi_k(t)$ into its Poisson zero-frequency part $\mathcal{P}(t) = M_u(MD) t^m$ and the periodic remainder $\mathcal{E}_d := S_m - \mathcal{P}$.

Theorem 4.2 (Explicit periodic residue). *$\mathcal{E}_d$ is C^∞ , one-periodic, mean zero, nonzero, with the absolutely convergent odd-frequency series*

$$\boxed{\mathcal{E}_d(t) = -\frac{(d+1)!}{(2\pi i)^{d+1}} \sum_{\substack{n \in \mathbb{Z} \\ n \text{ odd}}} \frac{\widehat{u}(n/2)}{n^{d+1}} e^{2\pi i n t}}, \quad \mathcal{E}_d\left(t + \frac{1}{2}\right) = -\mathcal{E}_d(t); \quad (4.2)$$

a cosine series for odd d , a sine series for even d . The canonical local defect is $\mathcal{D}_d(t) := t^{d+1} + \mathcal{E}_d(t) \in V_d$, and

$$V_d = \Pi_d \oplus \text{span}\{\mathcal{D}_d\}, \quad c_{\text{def}}(q) = \frac{\Lambda_d(q)}{A_d(1) (d + 1)!} \quad (4.3)$$

is the defect coordinate of an arbitrary synthesis.

Proof. In the Poisson expansion of S_m , even nonzero frequencies die (zero order $1 + \nu_2(Mn) > m$) and at odd n only the leading jet of $\widehat{u}(M\xi)$ survives; by the corpus's leading-jet formula it equals $-m! \widehat{u}(n/2)/n^m$, giving (4.2). Odd-frequency support gives the half-period antisymmetry, and non-vanishing follows from $\widehat{u}(n/2) \neq 0$ for odd n . $\mathcal{E}_d$ itself is *not* in V_d (else $t^{d+1} \in V_d$, contradicting $V_d \cap \mathbb{R}[t] = \Pi_d$); the corrected $\mathcal{D}_d = S_m - (\mathcal{P} - t^m)$ is, since $\mathcal{P} - t^m \in \Pi_d$. Its certificate value is that of the sequence k^{d+1} , i.e. $A_d(1)(d + 1)! \neq 0$, so $\mathcal{D}_d \notin \Pi_d$ and (4.3) follows from [theorem 3.5](#). $\square$

4.3 Cross-volume identities: the defect is the quadrature first-failure

The series (4.2) is not new to the corpus: it is, up to an explicit constant, the degree- $(m+1)$ error of the *shifted dyadic-comb quadrature* studied in Part I of the companion volume *Dyadic-Comb Frontiers*, where

$$Q_{m,p}(\theta) = 2^{-m} \sum_{k \in \mathbb{Z}} (2^{-m}(k + \theta))^p u(2^{-m}(k + \theta))$$

is exact for $p \leq m$ and first fails at $p = m + 1$ through the same odd-alias leading jets.

Theorem 4.3 (Defect–quadrature duality). *For every $d \geq 0$ and $t \in \mathbb{R}$,*

$$\mathcal{E}_d(t) = (-1)^{d+1} 2^{d(d+1)} \left(Q_{d,d+1}(t) - \int_{-1}^1 x^{d+1} u(x) dx \right). \quad (4.4)$$

Proof. Both sides are the odd-frequency alias series with leading jets of $\widehat{u}$ at the order- $(d+1)$ zeros: the companion volume’s first-failure theorem gives

$$Q_{d,d+1}(\theta) - \mu_{d+1} = - \left(-\frac{1}{2\pi i} \right)^{d+1} \frac{(d+1)!}{2^{d(d+1)}} \sum_{\ell \text{ odd}} \frac{\widehat{u}(|\ell|/2)}{\ell^{d+1}} e^{2\pi i \ell \theta},$$

and comparing with (4.2) term by term (the $(-1)^{d+1}$ absorbs the sign of $(-1)^{-(d+1)}$) gives (4.4). $\square$

Two families of consequences flow across the volumes.

Phase zeros. The quadrature’s parity-forced superconvergent phases become the trivial zeros of the defect: for even d (sine series) $\mathcal{E}_d(0) = \mathcal{E}_d(1/2) = 0$, matching the centered/half-shifted superconvergence at even level; for odd d the quarter-period zeros $\mathcal{E}_d(1/4) = \mathcal{E}_d(3/4) = 0$ match the quarter-shift superconvergence at odd level.

Special values. For odd d , evaluation at $t = 0$ gives, with the spectral Dirichlet series $D(s) = \sum_{q \text{ odd}} \widehat{u}(q/2) q^{-s}$ of the companion volume,

$$\mathcal{E}_d(0) = (-1)^{(d-1)/2} \frac{2(d+1)!}{(2\pi)^{d+1}} D(d+1) \in \mathbb{Q}, \quad (4.5)$$

rational because $D(2r)/\pi^{2r} \in \mathbb{Q}$ is proved there. The proved exact values $D(2), \dots, D(8)$ yield

$$\mathcal{E}_1(0) = \frac{1}{18}, \quad \mathcal{E}_3(0) = -\frac{23}{1350}, \quad \mathcal{E}_5(0) = \frac{1543}{119070}, \quad \mathcal{E}_7(0) = -\frac{1196113}{65063250}, \quad (4.6)$$

and $\mathcal{E}_d(1/2) = -\mathcal{E}_d(0)$ in each case. As a consistency check across independently computed tables: the companion volume’s exact phase-error experiments give $Q_{1,2}(0) - \mu_2 = \frac{1}{72}$ and $Q_{3,4}(0) - \mu_4 = -\frac{23}{5529600}$, and (4.4) converts them to $\mathcal{E}_1(0) = 2^2 \cdot \frac{1}{72} = \frac{1}{18}$ and $\mathcal{E}_3(0) = 2^{12} \cdot \left(-\frac{23}{5529600}\right) = -\frac{23}{1350}$ — in exact agreement with (4.5).

Research question 4.4 (Defect values at other dyadic rationals). *Beyond $t \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$, the values $\mathcal{E}_d(a/2^s)$ are character-twisted versions of $D(d+1)$ (finite combinations of $\sum_{q \text{ odd}} \widehat{u}(q/2) \chi(q) q^{-(d+1)}$ over 2^s -periodic characters χ). Do they admit closed forms in Fabius values, finite Thue–Morse sums, or q -binomial data, as the principal value (4.5) does? This refines the open “defect special values” question of the absorbed source and inherits the machinery of the companion volume’s first-defect chapter.*

5 The scale-ladder windows

The second construction inverts the refinement equation itself. It needs no alias analysis: one telescoping identity, iterated, produces polynomial tails whose truncation is exact.

5.1 The antiderivative train

Write $u_{h,c}(x) = u((x-c)/h)$ and $(Jf)(x) = \int_{-\infty}^x f$.

Proposition 5.1 (One-step identity). $Ju(x) = \sum_{k \geq 0} u\left(\frac{x-(2k+1)}{2}\right)$, a locally finite sum; hence $Ju_{h,c}(x) = h \sum_{k \geq 0} u\left(\frac{x-c-h(2k+1)}{2h}\right)$.

Proof. By the refinement derivative, $\frac{d}{dx}u\left(\frac{x-(2k+1)}{2}\right) = u(x-2k) - u(x-2k-2)$; the locally finite series telescopes to $u(x)$, and both sides vanish for $x \ll 0$. $\square$

Theorem 5.2 (Iterated train). For every $m \geq 1$,

$$J^m u_{h,c}(x) = h^m 2^{m(m-1)/2} \sum_{N \geq 0} p_m(N) u\left(\frac{x-c-h(2^m-1+2N)}{2^m h}\right), \quad (5.1)$$

locally finite, and the atom indexed by N has support $[c-h+2hN, c+h(2^{m+1}-1+2N)]$: its left endpoint is independent of the order m .

Proof. Induct with [theorem 5.1](#): integrating the N -th atom produces atoms at scale $2^{m+1}h$ and centers $c+h(2^{m+1}-1+2(N+2^m k))$, so the new multiplicity is $\sum_{k \geq 0} p_m(N' - 2^m k) = p_{m+1}(N')$ – exactly the recurrence (2.3) – while the prefactor gains $2^m h$. The support bookkeeping is immediate from $\text{supp } u_{\rho,c'} = [c' - \rho, c' + \rho]$. $\square$

The weights are thus the Green sequence of [theorem 2.2](#): repeated *continuous* antidifferentiation of the refinement derivative and the *discrete* inversion of the Thue–Morse difference stencil are the same computation.

5.2 Polynomial tails and the finite window

Lemma 5.3 (Tail polynomial). For $n \geq 0$ and $x \geq c+h$, $J^{n+1}u_{h,c}(x) = \frac{h^{n+1}}{n!} G_n\left(\frac{x-c}{h}\right)$, with G_n the moment Appell family (0.1).

Proof. $J^{n+1}u_{h,c}(x) = \frac{1}{n!} \int (x-t)^n u\left(\frac{t-c}{h}\right) dt$ over the full support once $x \geq c+h$; substitute $t = c + hz$. $\square$

Combining [theorems 5.2](#) and [5.3](#) at order $m = n+1$ gives the locally finite half-line identity

$$G_n\left(\frac{x-c}{h}\right) = n! 2^{n(n+1)/2} \sum_{N \geq 0} p_{n+1}(N) u\left(\frac{x-c-h(2^{n+1}-1+2N)}{2^{n+1}h}\right) \quad (x \geq c+h), \quad (5.2)$$

and the support-edge invariance makes one truncation index serve all degrees simultaneously:

Theorem 5.4 (Exact finite polynomial window). Let $a < b$, $L = b-a$, $h > 0$, $K = \lceil L/(2h) \rceil$, and for $n \geq 0$

$$\Psi_{n,h}(x) = n! 2^{n(n+1)/2} \sum_{N=0}^K p_{n+1}(N) u\left(\frac{x-a-h(2^{n+1}-2+2N)}{2^{n+1}h}\right). \quad (5.3)$$

Then $\Psi_{n,h}(x) = G_n\left(\frac{x-a+h}{h}\right)$ for all $x \in [a, b]$; in particular $\{\Psi_{0,h}, \dots, \Psi_{d,h}\}$ restricts to a monic triangular basis of Π_d on $[a, b]$, using $(d+1)(K+1)$ atoms at the scales $2h, 4h, \dots, 2^{d+1}h$.

Proof. Take $c = a-h$ in (5.2), so $x \geq c+h$ on all of $[a, b]$. Every omitted atom $N \geq K+1$ has left support endpoint $a-2h+2hN \geq a+2hK \geq b$, hence vanishes on $[a, b]$ (at a boundary touch the argument is -1 and $u(-1) = 0$). $\square$

Corollary 5.5 (Two atoms per degree). *For $h \geq L/2$, $K = 1$ and $p_{n+1}(0) = p_{n+1}(1) = 1$: each block is a pair of equal-amplitude atoms, and with the canonical choice $h = L/2$,*

$$\boxed{\Psi_n(x) = n! 2^{n(n+1)/2} \left[u\left(\frac{x-a-(2^n-1)L}{2^n L}\right) + u\left(\frac{x-a-2^n L}{2^n L}\right) \right] = G_n\left(1 + \frac{2(x-a)}{L}\right)} \quad (5.4)$$

on $[a, b]$. Every $p \in \Pi_d$ therefore uses at most $2d + 2$ raw atoms; the case $n = 0$ is the partition of unity (1.2).

5.3 Exact coefficients: the Appell transform

Theorem 5.6 (Coefficient transform). *Let $p \in \Pi_d$, $c = a - h$, $\tilde{p}(y) = p(c + hy)$. The unique expansion $\tilde{p} = \sum_{n \leq d} b_n G_n$ has*

$$b_n = [y^n] M_u(D)^{-1} \tilde{p} = \sum_{r=0}^{d-n} \frac{\gamma_r h^{n+r}}{n! r!} p^{(n+r)}(c) = \sum_{r=0}^{d-n} \binom{n+r}{r} \gamma_r c_{n+r}, \quad (5.5)$$

the last form for $\tilde{p} = \sum_j c_j y^j$; equivalently the monic-triangular descending recurrence $b_n = c_n - \sum_{k>n} b_k \binom{k}{n} \mu_{k-n}$ holds. Then

$$p(x) = \sum_{n=0}^d b_n \Psi_{n,h}(x) \quad (a \leq x \leq b), \quad (5.6)$$

with raw atom amplitudes $\alpha_{n,N} = n! 2^{n(n+1)/2} b_n p_{n+1}(N)$ – in the canonical two-atom form, both atoms of block n share the amplitude $\alpha_n = n! 2^{n(n+1)/2} b_n$. For monomials the coefficients are closed Bernoulli–Bell forms: $p(x) = (x-a)^m$ has $b_n = h^m \binom{m}{n} \mathcal{A}_{m-n}(-1)$. Everything is rational for rational data, affine-covariant, and computable in $\mathcal{O}(d^2 + dK)$ exact operations.

Proof. $G_n = M_u(D)y^n$ makes $M_u(D)$ triangular-invertible on Π_d ; apply $M_u(D)^{-1}$ to the expansion, extract coefficients, and use (5.3). For the monomial form: with $c = a - h$, $(x-a)^m = h^m(y-1)^m$, Appell operators commute with shifts, so $M_u(D)^{-1}(y-1)^m = \mathcal{A}_m(y-1) = \sum_n \binom{m}{n} \mathcal{A}_{m-n}(-1) y^n$ by the Appell binomial theorem. $\square$

Example 5.7 (Worked quartic). For $p(x) = 3x^4 - 2x^3 + 5x - 7$ on $[0, 1]$ with $h = \frac{1}{2}$ ($c = -\frac{1}{2}$): $\tilde{p}(y) = \frac{3}{16}y^4 - y^3 + \frac{15}{8}y^2 + y - \frac{145}{16}$, the Appell coordinates are $(b_0, \dots, b_4) = (-\frac{2084}{225}, \frac{4}{3}, \frac{7}{4}, -1, \frac{3}{16})$ (re-verified exactly for this consolidation via the triangular recurrence), the block amplitudes are $(\alpha_0, \dots, \alpha_4) = (-\frac{2084}{225}, \frac{8}{3}, 28, -384, 4608)$, and

$$p(x) = \sum_{n=0}^4 \alpha_n \left[u\left(\frac{x-(2^n-1)}{2^n}\right) + u\left(\frac{x-2^n}{2^n}\right) \right] \quad (0 \leq x \leq 1), \quad (5.7)$$

a ten-atom exact window. The absorbed source verifies the identity symbolically (residual exactly zero) and independently by FFT inversion of a 44-factor sinc product (10^{-8} -level numerical residuals measuring only the Fourier grid); see [figure 1](#).

5.4 Structure: Chebyshev system, extension operator, collars

Proposition 5.8 (Derivative ladder, Wronskian, interpolation). *On $[a, b]$, $\Psi_{n,h}^{(k)} = n^k h^{-k} \Psi_{n-k,h}$; the Wronskian is constant, $W(\Psi_{0,h}, \dots, \Psi_{d,h}) = h^{-d(d+1)/2} \prod_{n \leq d} n!$; and for distinct nodes $x_0, \dots, x_d \in [a, b]$,*

$$\det[\Psi_{j,h}(x_i)] = h^{-d(d+1)/2} \prod_{i < j} (x_j - x_i). \quad (5.8)$$

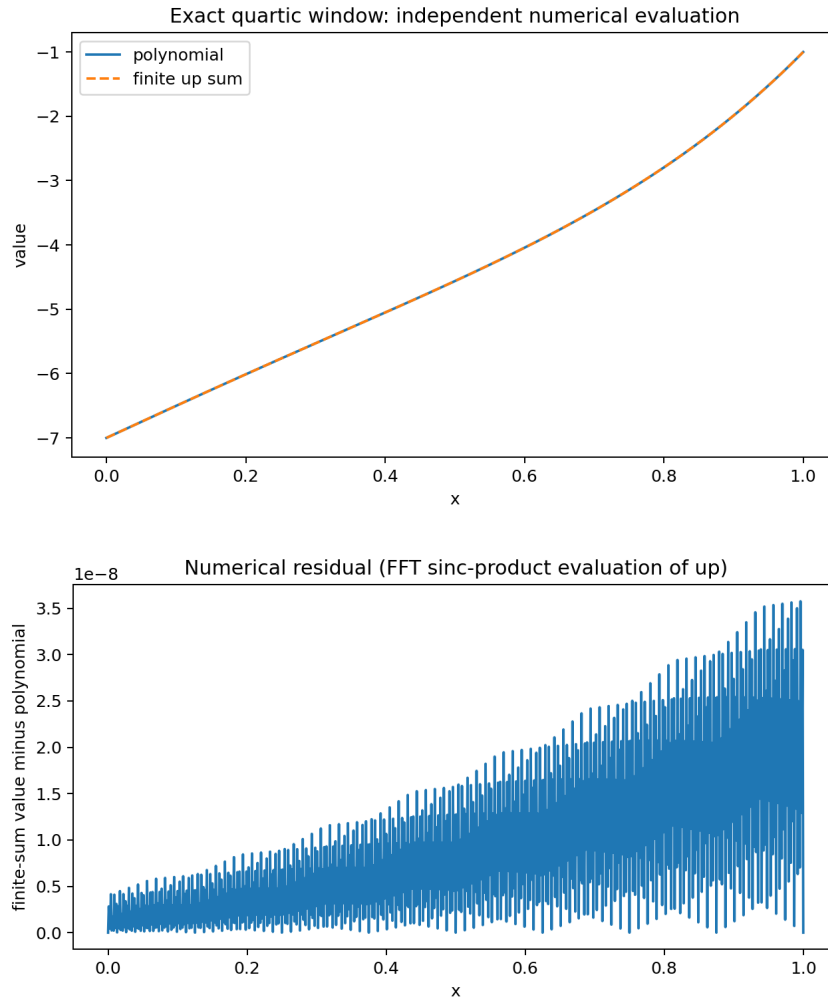


Figure 1: The worked quartic (5.7): the ten-atom window (top, visually indistinguishable from the polynomial on $[0, 1]$) and the residual of the independent FFT check (bottom); the exact symbolic residual is zero.

The block family is thus an extended complete Chebyshev system on every subinterval, nonsingular for interpolation at every node set.

Proof. On the interval the blocks are the monic polynomials $G_n((x - a + h)/h)$; a monic triangular change of basis has determinant one, the monomial Wronskian is $\prod n!$, each derivative row scales by h^{-k} , and the interpolation determinant is the Vandermonde in $y_i = (x_i - a + h)/h$. $\square$

Proposition 5.9 (Extension operator and support). *The map $E_{d,a,b,h}: p \mapsto \sum_{n,N} \alpha_{n,N}(p) u((x - C_{n,N})/2^{n+1}h)$, with centers $C_{n,N} = a + h(2^{n+1} - 2 + 2N)$, is a linear injection $\Pi_d \rightarrow C_c^\infty(\mathbb{R})$ that restricts to the identity on $[a, b]$ — an explicit compactly supported smooth extension operator for polynomial data, flat at its outer support endpoints, with*

$$\text{supp } E_{d,a,b,h} p \subseteq [a - 2h, a + h(2^{d+2} - 2 + 2K)] \subset [a - 2h, b + 2^{d+2}h]. \quad (5.9)$$

Theorem 5.10 (Collar–atom tradeoff). *For any collar $\rho > 0$, the choice $h = \rho/2^{d+2}$ gives an exact window supported in $[a - \rho, b + \rho]$ with at most $(d + 1)(1 + \lceil 2^{d+1}L/\rho \rceil)$ atoms. Conversely the sparse $h = L/2$ choice has linear atom count $2d + 2$ but an exponentially long one-sided scale ladder, $\text{supp} \subseteq [a - L, a + 2^{d+1}L]$. A reflected anticausal train (right-tail integration J_+) mirrors the overhang, and orientations may be mixed degree by degree, since any sequence of monic degree- n blocks remains a triangular basis.*

Proof. $2h = \rho/2^{d+1} \leq \rho$ bounds the left overhang and (5.9) the right one; the count is $(d + 1)(K + 1)$. The reflection statement is $J_+ u(x) = \sum_{k \geq 0} u(\frac{x+2k+1}{2})$ by evenness. $\square$

Remark 5.11 (Conditioning). The level normalization $n! 2^{n(n+1)/2}$ grows superexponentially and the paired atoms are sampled near flat support edges, so the polynomial emerges from cancellation among large terms: construct coefficients exactly, evaluate in extended precision, sum blockwise with compensation. The quasipolynomial law (2.4) shows why shrinking h does not uniformly worsen coefficients (normalized weights grow like N^n while the Appell coordinates carry powers of h): a nontrivial conditioning-optimal h should exist (theorem 8.5). The same warning applies to the endpoint basis of section 3, whose minimal atom count also comes at the price of large signed coefficients.

6 Dictionary versus ladder, and the minimal atom count

The two constructions solve the same problem with complementary economies:

	Common-scale dictionary (section 3)	Scale ladder (section 5)
Scales used	one: $2^d h$	$d + 1$: $2h, 4h, \dots, 2^{d+1}h$
Atoms for Π_d , one cell	$d + 2$ (optimal for fixed dictionaries)	$2d + 2$ (two per degree, equal amplitudes per block)
Coefficients	recurrence residual / $H_d(D)$ symbol, $\mathcal{O}(d^2)$	triangular Appell transform, $\mathcal{O}(d^2)$
Uniqueness	unique in the endpoint basis	unique in the block basis
Support overhang	$[a - (d + 1)h, b + (2M - 1)h]$, knob J	collar ρ at will via $h = \rho/2^{d+2}$, knob h
Certificate / defect	one-scalar Thue–Morse test; explicit $\mathcal{D}_d$	exactness by construction (blocks are G_n)
Discrete engine	alias annihilator $A_d = \prod [2^m]_z$	Green inverse $1/\Theta_m$, weights $p_m(N)$

The two discrete engines are the two directions of one convolution identity (Θ against its causal inverse), and the two coefficient calculi are the two members of the Appell pair (0.1). The comparison also settles more of the minimal-atom question than either source alone.

Theorem 6.1 (Minimal atom counts). *Let N_d be the least integer such that every real polynomial of degree at most d admits a representation (1.1) on every nondegenerate interval with at most N_d atoms. Then*

$$N_0 = 2, \quad \boxed{N_d \leq d + 2} \quad (d \geq 0), \quad (6.1)$$

the upper bound achieved by the single fixed endpoint dictionary of theorem 3.7; and every fixed dictionary whose span contains Π_d has at least $d + 2$ atoms (theorem 3.6).

Proof. $N_0 \geq 2$ is the strict monotonicity of section 1; $N_0 \leq 2$ is the partition (1.2). For general d the endpoint basis represents every $P \in \Pi_d$ with the same $d + 2$ atoms. $\square$

The bound $N_d \leq d + 2$ sharpens the ladder count $2d + 2$ of theorem 5.5 — one of the dividends of consolidating the two sources. What remains open is whether *adaptivity* can beat the dictionary:

Conjecture 6.2 (Target-adaptive rigidity). *For Lebesgue-almost every $P \in \Pi_d$ there is no identity (1.1) with $N \leq d + 1$ atoms, even when amplitudes, shifts, and scales all depend on P ; hence $N_d = d + 2$. The fixed-dictionary case is theorem 3.6; the adaptive case replaces a finite union of proper subspaces by an uncountable parametrized family and appears to require a rigidity theorem for cancellation among nonanalytic affine copies of u . Within the causal-ladder class (degree-independent left support edge, exact degree- n polynomial tails), at least two atoms per active level are conjectured necessary for generic targets, making $2d + 2$ minimal in that class.*

Remark 6.3 (Strang–Fix reconciliation). There is no tension with the classical no-go: a single fixed-scale integer-shift u -system reproduces only constants globally (theorem 3.1 at $\nu_2(\rho) = 0$). Both constructions bypass it lawfully — the dictionary by paying scale ($\rho = 2^d$ restores order d globally before localizing), the ladder by paying scales *and* asking only for a window (boundary truncation on a bounded interval). The classical UP_n polynomial inclusions of Rvachev theory [3, 5] are thereby made explicit in two normal forms.

7 Extensions

7.1 Odd overscaling

For $\rho = 2^d q$ with q odd, reproduction order stays d (theorem 3.1) while the alias argument yields the larger annihilator $A_{\rho,d}(z) = \prod_{r=0}^d (z^{\rho/2^r} - 1)/(z - 1)$ of degree $2\rho - q - d - 1$, from the coset multiplicities $1 + \min\{\nu_2(j), d\}$; the one-cell dimension is $\leq q + d + 1$, conjecturally with equality (theorem 8.2): each odd factor buys $q - 1$ extra *defect* directions and no polynomial order.

7.2 Tensor products and multivariate compression

On a box in $\mathbb{R}^s$, both constructions tensorize: coordinatewise degrees $(d_1, \dots, d_s)$ cost $\prod_j (d_j + 2)$ dictionary atoms or $\prod_j 2(d_j + 1)$ ladder atoms, exactly on the tensor polynomial space. For *total* degree d the tensor counts far exceed $\binom{d+s}{s}$; a multivariate alias-ideal compression (the analogue of theorem 3.4 for $\mathbb{Z}^s$ -cosets) is an attractive frontier problem.

7.3 Hermite jet stitching and analytic data

The derivative ladder of theorem 5.8 exposes all endpoint jets of the blocks, so finite Hermite systems stitch different polynomial plateaus on disjoint intervals into one compactly supported C^∞ function with automatic smooth transitions — an atomic-function analogue of splines with infinitely smooth knots. For f analytic near $[a, b]$, truncating the Appell expansion of $f(c + hy)$ at degree d converts any polynomial-approximation remainder into a finite-atom approximation bound: the exact theory doubles as a constructive approximation scheme.

7.4 Base b and q -ary ladders

Both engines are dyadic only through the zero set of $\widehat{u}$. For an integer base $b \geq 2$, the compactly supported density with transform $\prod_{j \geq 0} \text{sinc}_\pi(\xi/b^j)$ supports the same two constructions – dictionary scale b^d , and ladders weighted by b -ary restricted partition numbers $[z^N] \prod_{j < m} (1 - z^{b^j})^{-1}$ whenever the derivative refinement has an integral translation stencil – with a generalized Prouhet digit character as certificate. The dyadic case is the most arithmetic because the reciprocal stencil is the Thue–Morse block.

7.5 Positive synthesis and dual functionals

Sign-changing polynomials force signed amplitudes, but for strictly positive targets, nonnegative-amplitude synthesis in an overcomplete dictionary is a finite-dimensional cone-membership problem after fixing scale and interval (theorem 8.8). Compactly supported dual functionals λ_n with $\lambda_n(\Psi_m) = \delta_{nm}$ – ideally finite dyadic samples or moments of u – would upgrade the block basis to a local projection and quadrature calculus (theorem 8.9).

8 Conjectures, problems, and the Lean program

Besides theorem 6.2 and theorem 4.4:

Conjecture 8.1 (Central consecutive basis). *The consecutive central block $\{\psi_0, \dots, \psi_{d+1}\}|_{[0,1]}$ is a basis of V_d for every d ; equivalently the integer determinant Δ_d^{cen} of its reduction to the endpoint basis is nonzero. Exact values through $d = 12$ support this:*

$$1, 1, 3, -1, 255, -6145, 524287, -62914561, 17716740095, -9620726743041, \\ 11047892835893247, -25076042725198921729, 113484369541461161541631,$$

a sequence with visibly rich divisibility (note $255 = 2^8 - 1$, $524287 = 2^{19} - 1$) inviting a product, resultant, or cyclotomic interpretation (theorem 8.7). Nonvanishing is an algebraic claim only; some central coordinates are worse conditioned than endpoint ones.

Conjecture 8.2 (Odd overscaling dimension). *For $\rho = 2^d q$, q odd, the one-cell local space has dimension exactly $q + d + 1$.*

Conjecture 8.3 (Bit growth). *The lowest-terms numerators and denominators of the endpoint coefficients of t^d have bit length $\mathcal{O}(d^2)$. (Plausible since $\log A_d(1) = \Theta(d^2)$ and all operator series are Bernoulli-rational; cancellation makes a proof nontrivial.)*

Conjecture 8.4 (Support–atom Pareto optimality). *For fixed d and collar ratio $\rho/L \rightarrow 0$, any causal positive-scale construction with uniformly controlled block condition numbers needs $\Omega_d(L/\rho)$ atoms; the bound of theorem 5.10 is optimal up to a degree-dependent constant.*

Conjecture 8.5 (Conditioning-optimal scale). *Natural coefficient and evaluation condition functionals of the ladder have a finite minimizer in $h > 0$, unique up to a short list of adjacent K -regimes.*

Research question 8.6 (Condition-optimal $(d+2)$ -basis). *Among all $(d+2)$ -subsets of the active dyadic dictionary, which basis minimizes a chosen stability functional (L^∞ coefficient norm over unit-norm targets, smallest sampled singular value, Lebesgue-type constant, support overhang)?*

Research question 8.7 (Central determinant sequence). *Find a recurrence, product formula, or resultant interpretation for Δ_d^{cen} and explain its Mersenne-type prime content.*

Research question 8.8 (Positive synthesis). *Characterize the polynomials representable with nonnegative amplitudes in an overcomplete up-dictionary.*

Problem 8.9 (Dual functionals). *Construct compactly supported duals of the block (or endpoint) basis from finite dyadic samples or moments of u , and derive the induced local projection and quadrature formulas.*

Problem 8.10 (Weight arithmetic). *Expand $p_m(N)$ and the normalized monic weights in bases adapted to Gaussian q -binomial coefficients and Prouhet cancellation; make the lower-order quasipolynomial coefficients explicit as Bernoulli polynomials on residue classes modulo 2^{m-1} , and connect them to Mahler-type functional equations.*

Lean formalization roadmap

In dependency order, all finite or formal-power-series layers first: (1) the telescoping identity of [theorem 5.1](#) and the binary-partition induction (5.1); (2) support bookkeeping and the truncation theorem [theorem 5.4](#); (3) finite-dimensional Appell inversion ([theorem 5.6](#)); (4) polynomial-weighted Poisson summation and the global synthesis [theorem 3.3](#) (the analytic core, using the already formalized integer zero orders); (5) root-of-unity comb derivatives and the alias multiplicities of [theorem 3.4](#); (6) the q -integer factorization (2.5); (7) the dimension argument of [theorem 3.5](#) via the existing non-elementarity theorem; (8) recurrence elimination, the Thue–Morse certificate, and the formal cancellation proof of the H_d symbol (3.12). Steps (1)–(3) and (5)–(8) are finite algebra; only (4) needs Schwartz-class Poisson summation.

A Low-order operator data

Truncations of the fast symbol (acting on degrees shown):

$$H_0 = 1, \quad H_1 = 1 + \mathcal{O}(z^2), \quad H_2 = 1 - \frac{5}{36}z^2 + \mathcal{O}(z^4), \quad H_3 = 1 - \frac{13}{72}z^2 + \mathcal{O}(z^4), \quad H_4 = 1 - \frac{2}{9}z^2 + \frac{857}{32400}z^4 + \mathcal{O}(z^6);$$

low annihilator coefficients $(a_0, \dots, a_{d+1})$: $(1, 1, 0)$, $(1, 2, 2, 2)$, $(1, 3, 5, 7, 8)$, $(1, 4, 9, 16, 24, 32)$, $(1, 5, 14, 30, 54, 86, 126)$ for $d = 1, \dots, 5$; and the formula sheet of the ladder (all objects rational):

cumulants	$\kappa_{2m} = B_{2m}/(2m(1 - 4^{-m}))$, odd zero
reciprocal MGF	$\gamma_0 = 1, \quad \gamma_n = -\sum_r \binom{n-1}{r-1} \kappa_r \gamma_{n-r}$
blocks	$\Psi_{n,h} = n! 2^{n(n+1)/2} \sum_{N \leq K} p_{n+1}(N) u((x - C_{n,N})/2^{n+1}h)$
centers/scales	$C_{n,N} = a + h(2^{n+1} - 2 + 2N)$, $2^{n+1}h$, $K = \lceil (b-a)/2h \rceil$
coordinates	$b_n = \sum_r \gamma_r h^{n+r} p^{(n+r)}(a-h)/(n! r!)$
amplitudes	$\alpha_{n,N} = n! 2^{n(n+1)/2} b_n p_{n+1}(N)$
support	$[a - 2h, a + h(2^{d+2} - 2 + 2K)]$

B Provenance of the absorbed sources

Source (directory; SHA-256 of main .tex)	Contribution
<i>Exact Polynomial Synthesis by Finite Rvachev Up-Function Dictionaries</i> (Rvachev_Up_Polynomial_Representation_Package; a574b30dadbc. . .)	The common-scale construction of sections 3 and 4 : scale classification, Poisson collapse, deconvolution synthesis, alias nullspace and q -integer annihilator, $\dim V_d = d+2$, endpoint basis and fixed-dictionary minimality, output-sensitive H_d algorithm, multi-cell theorem, Thue–Morse certificate, canonical defect and Fourier residue, central-basis determinants, odd overscaling, monomial examples and exact data files.
<i>Exact Polynomial Windows from Finite Sums of Shifted and Scaled Rvachev up Functions</i> (rvachev_up_polynomial_representation; 98fe2e5fd256. . .)	The ladder construction of section 5 : antiderivative train, binary-partition weights and Thue–Morse Green pair, quasipolynomial structure, tail polynomials, finite-window and two-atom theorems, Appell coefficient transform with Bernoulli–Bell closed forms, Wronskian/Chebyshev structure, extension operator, collar-atom tradeoff, anticausal mixing, conditioning analysis, worked quartic with figures, classical Rvachev bibliography.

The exact SHA-256 hashes of the two absorbed .tex sources are recorded in `assets/SHA256S UMS-absorbed.txt`. *What was deduplicated*: the corpus background (normalization, probability model, MGF/cumulants, Fourier zeros), the Appell calculus, the Thue–Morse products, rationality and affine covariance, tensor products, extension-operator remarks, conditioning warnings, and the Lean roadmaps, each of which both sources develop. *What is new to the consolidation*: the unified Appell-pair and Green-pair framing of [section 2](#), the comparison table and the sharpened bound $N_d \leq d+2$ of [theorem 6.1](#), and the cross-volume identities of [section 4.3](#) — the defect–quadrature duality (4.4) with its exact special values (4.6), cross-validated against the companion volume’s independently computed phase-error table. Beyond notation, no mathematical content of either source was discarded. Both sources audited the repository corpus (67-path inventory) and agree that the explicit finite constructions were absent from it; that corpus-relative novelty standard is inherited here.

References

- [1] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.
- [2] V. L. Rvachev and V. A. Rvachev, A certain finite function, *Dopovidi Akademii Nauk Ukrains’koi RSR, Series A* (1971), 705–707, 764.
- [3] V. L. Rvachev and V. A. Rvachev, On the representation of polynomials by functions with compact support, *Mathematical Physics*, No. 11, Naukova Dumka, Kiev, 1972, 126–129.
- [4] V. A. Rvachev, On approximation by means of the function $\text{up}(x)$, *Doklady Akademii Nauk SSSR* **233** (1977), no. 2, 295–296.
- [5] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.
- [6] J. Arias de Reyna, Definición y estudio de una función indefinidamente diferenciable de soporte compacto, *Rev. Real Acad. Ciencias Madrid* **76** (1982), 21–38; English translation arXiv:1702.05442.

- [7] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Article A51.
- [8] G. Strang and G. Fix, A Fourier analysis of the finite element variational method, in *Constructive Aspects of Functional Analysis*, C.I.M.E., 1973, pp. 793–840.
- [9] R.-Q. Jia, Shift-invariant spaces and linear operator equations, *Israel Journal of Mathematics* **103** (1998), 259–288.
- [10] C. de Boor and A. Ron, Fourier analysis of the approximation power of principal shift-invariant spaces, *Constructive Approximation* **8** (1992), 427–462.
- [11] C. de Boor, R. A. DeVore, and A. Ron, Approximation from shift-invariant subspaces of $L_2(\mathbb{R}^d)$, *Transactions of the American Mathematical Society* **341** (1994), 787–806.
- [12] M. Charina and V. Yu. Protasov, Analytic functions in local shift-invariant spaces and analytic limits of level dependent subdivision, *Journal of Fourier Analysis and Applications* **27** (2021), Article 45.
- [13] S. L. Lee, Approximation of Gaussian by scaling functions and biorthogonal scaling polynomials, *Bull. Malays. Math. Sci. Soc. (2)* **32** (2009), no. 3, 261–282.
- [14] V. F. Kravchenko, O. V. Kravchenko, Ya. Yu. Kononov, V. I. Pustovoi, and D. V. Churikov, Atomic, WA-systems, and R-functions applied in modern radio physics problems: Part III, *Journal of Communications Technology and Electronics* **60** (2015), 707–736.
- [15] S. Roman, *The Umbral Calculus*, Academic Press, 1984.
- [16] G. E. Andrews, *The Theory of Partitions*, Cambridge University Press, 1998 reprint.
- [17] L. Comtet, *Advanced Combinatorics*, Reidel, 1974.
- [18] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [19] V. Reshetnikov, *The Fabius Function and Rvachev’s Up-Function*, primary exposition in the ProveIt repository, Analysis/FabiusFunction/docs, snapshot 28 August 2026.
- [20] V. Reshetnikov, *Dyadic-Comb Frontiers for the Fabius–Rvachev System*, consolidated frontier volume, drafts/inverse-and-sampling/Dyadic_Comb_Frontiers/, 28 August 2026.
- [21] V. Reshetnikov, *Inverse and Sampling Frontiers for the Fabius–Rvachev System*, consolidated frontier volume, drafts/inverse-and-sampling/Inverse_and_Sampling_Frontiers/, 2026.